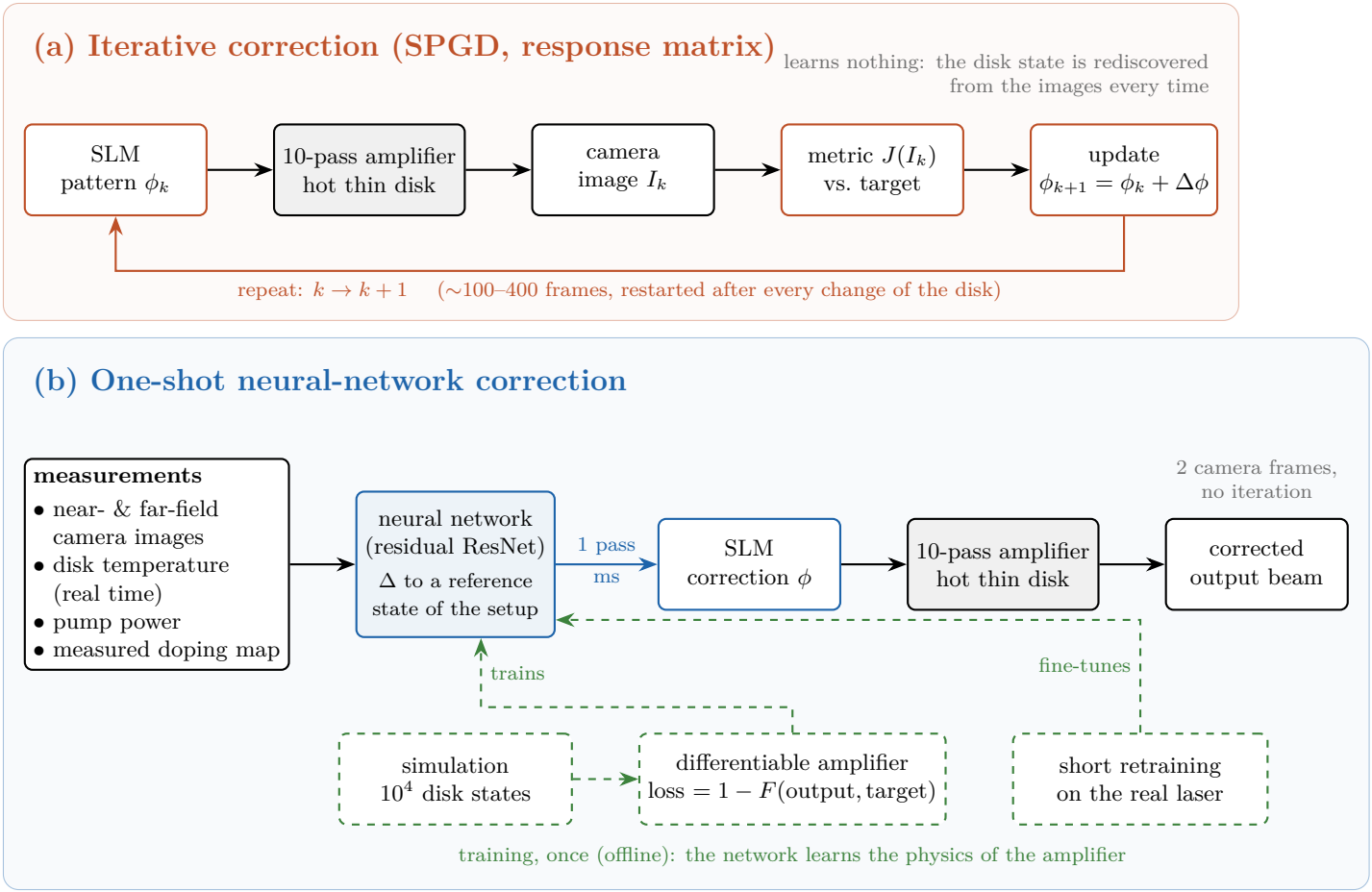

(a) Iterative correction (SPGD, response matrix)
learns nothing: the disk state is rediscovered from the images every time
SLM pattern ϕ_k
10-pass amplifier hot thin disk
camera image I_k
metric $J(I_k)$ vs. target
update $\phi_{k+1} = \phi_k + \Delta\phi$
repeat: $k \to k+1$ (∼100–400 frames, restarted after every change of the disk)
(b) One-shot neural-network correction
measurements
• near- & far-field camera images
• disk temperature (real time)
• pump power
• measured doping map
neural network (residual ResNet)
Δ to a reference state of the setup
1 pass
ms
SLM correction ϕ
10-pass amplifier hot thin disk
corrected output beam
2 camera frames, no iteration
trains
fine-tunes
simulation 10^4 disk states
differentiable amplifier loss $= 1 - F(\text{output}, \text{target})$
short retraining on the real laser
training, once (offline): the network learns the physics of the amplifier